

PLANAR MACHINE

SHIVAM RAJ
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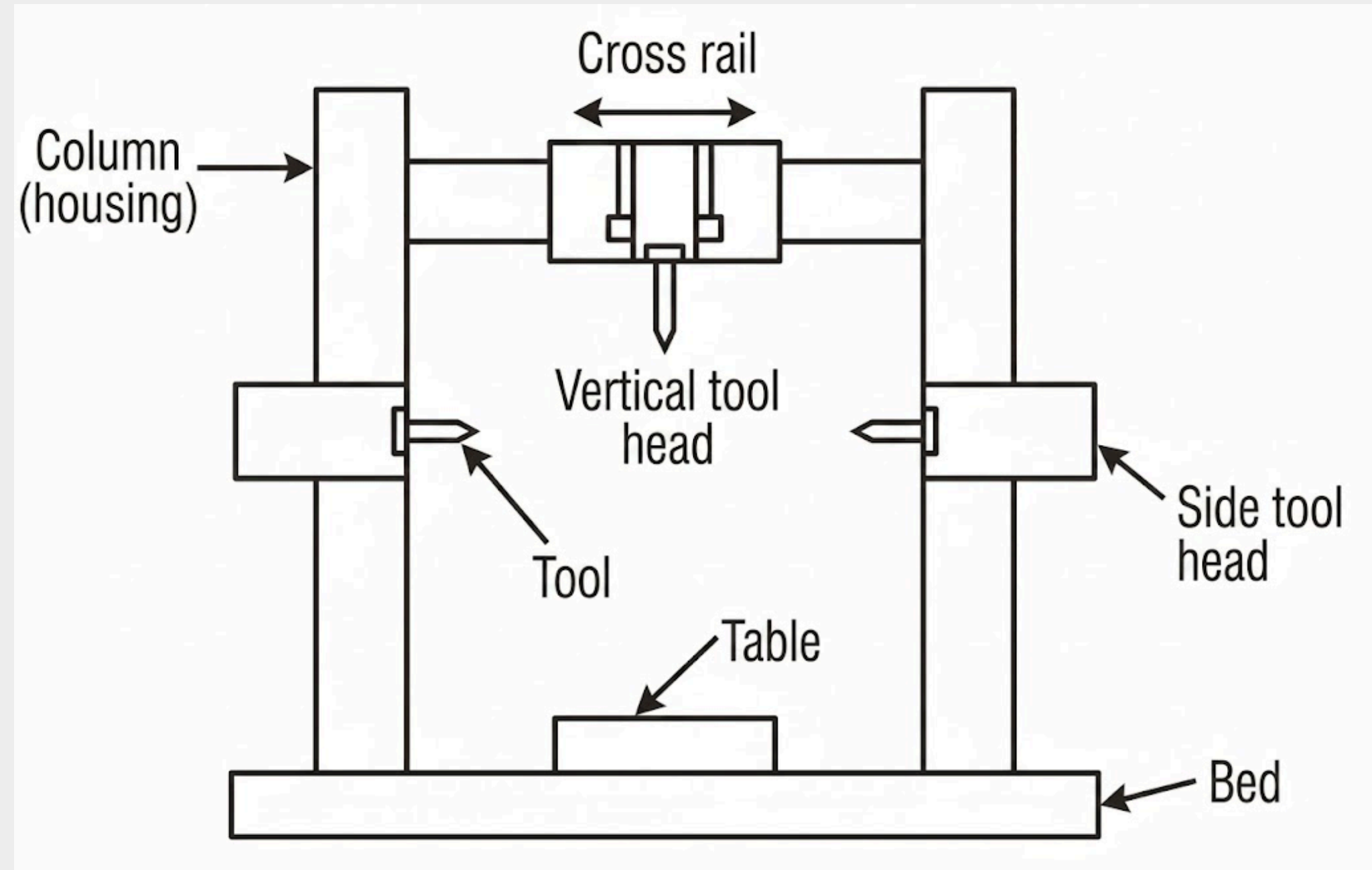
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INTRODUCTION

- Planer is machine that use to generate accurate flat surfaces and cutting slots.
- It is similar to a shaper, but it is larger than shaper machine.
- The entire workpiece moving beneath the cutter.
- The work table is moved back and forth on the bed beneath the cutting head either by mechanical means, such as a rack and pinion gear, or by a hydraulic system.

CONSTRUCTION

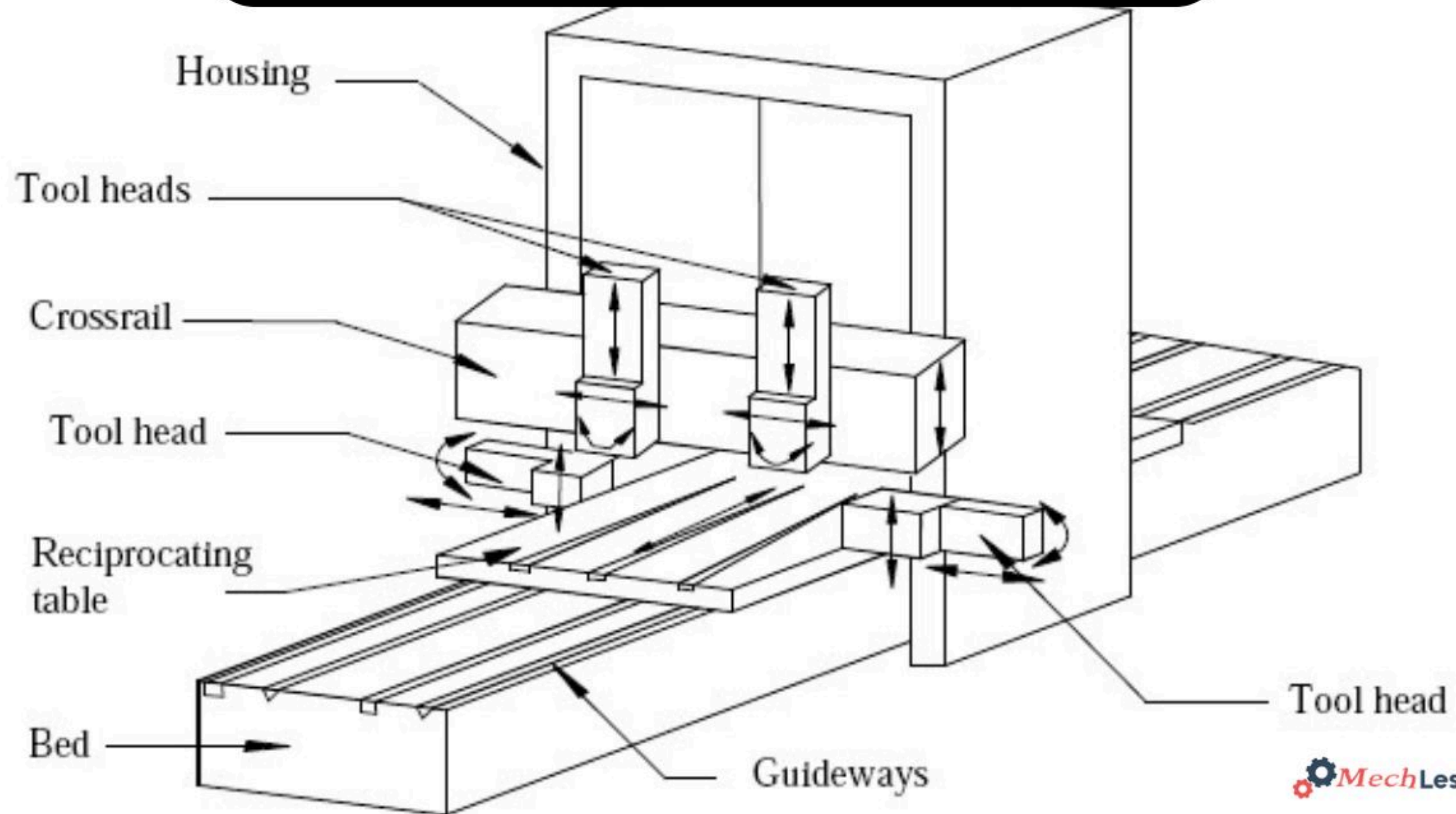
- **BED**
- **TABLE**
- **COLUMN**
- **CROSSRAIL**
- **TOOL HEAD**



COMPONENTS DESCRIPTION

- **Bed – Base of the machine that supports all components.**
- **Table – Holds the workpiece and moves back and forth.**
- **Column – Vertical structures that support the cross rail.**
- **Cross Rail – Connects columns and carries the tool head horizontally.**
- **Tool Head – Holds and controls the cutting tool for machining.**

DIAGRAM OF PLANER MACHINE



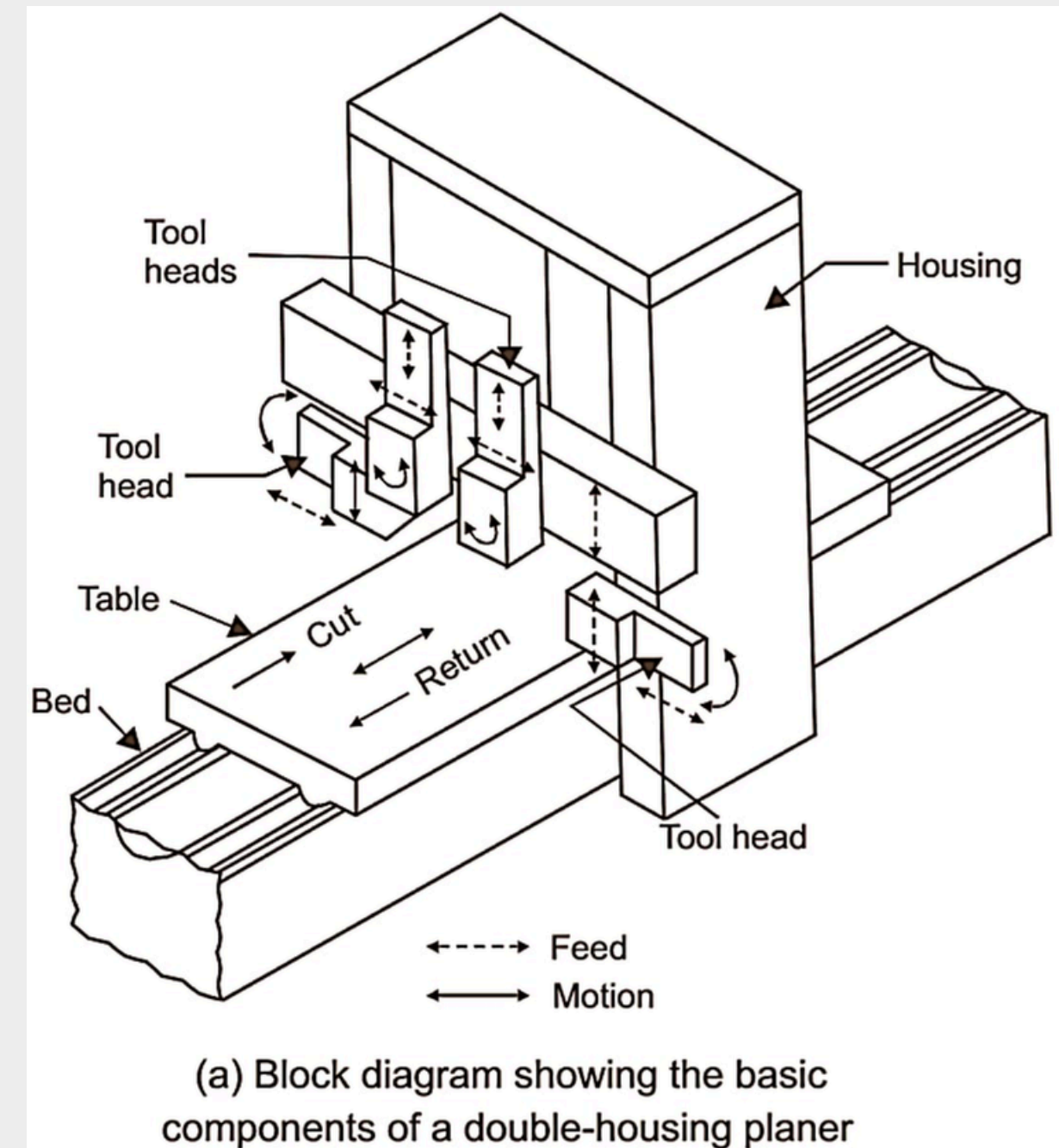
MECHANISM OF WORKING

- The workpiece is fixed on the table, and the table reciprocates (moves back and forth).
- The cutting tool remains stationary except for feed movement.
- During the forward stroke, cutting takes place (material is removed).
- During the return stroke, no cutting occurs (idle stroke).
- The table movement is driven by mechanical (rack & pinion) or hydraulic system.
- After each stroke, the tool is given a small feed (sideways or vertical) for the next cut.
- This process continues until the required flat surface or slot is obtained.

TYPES OF PLANAR MACHINE

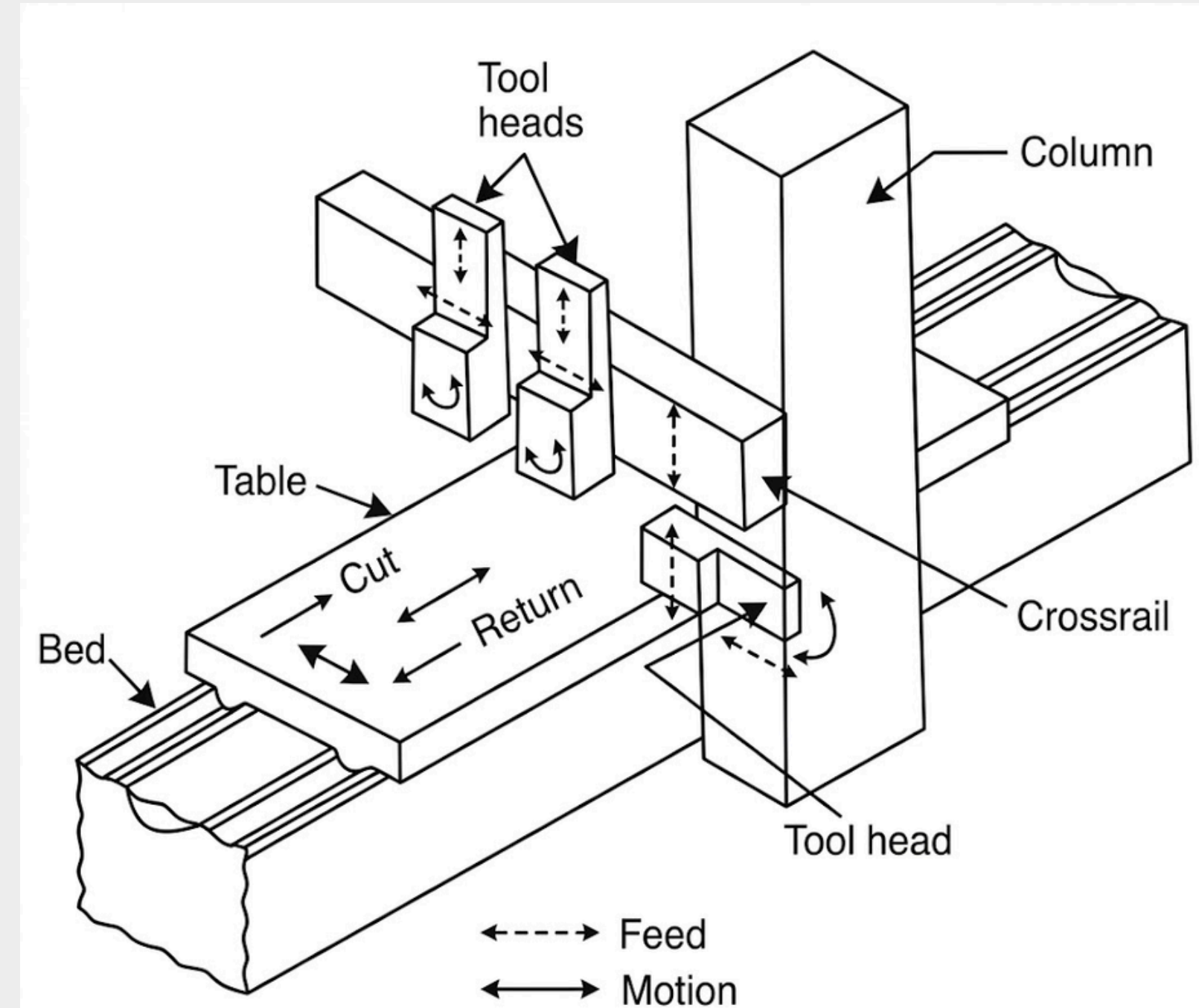
DOUBLE HOUSING PLANAR

- It is very old system machine.
- Massive bed on which worktable reciprocates
- A planer having two housings to support the cross rail, with two heads on the cross rail.
- Two vertical columns on which two tool head slides
- Cross rail fitted between two columns and carries one or two heads slides horizontally.



OPEN SIDE PLANAR

- It consist of one vertical column may be mounted on the cross rail
- column and cross rail consist of single and double tool head
- Used for the machining of wide work pieces



(b) Block diagram of an open-side planer

DIVIDE TYPE PLANAR

- It also called tandem planer.
- Consist of two worktable.
- Used for continuous mass production.
- On one table the work piece is being machined and on the other table work piece is on standby.
- Two table are also join together when needed.

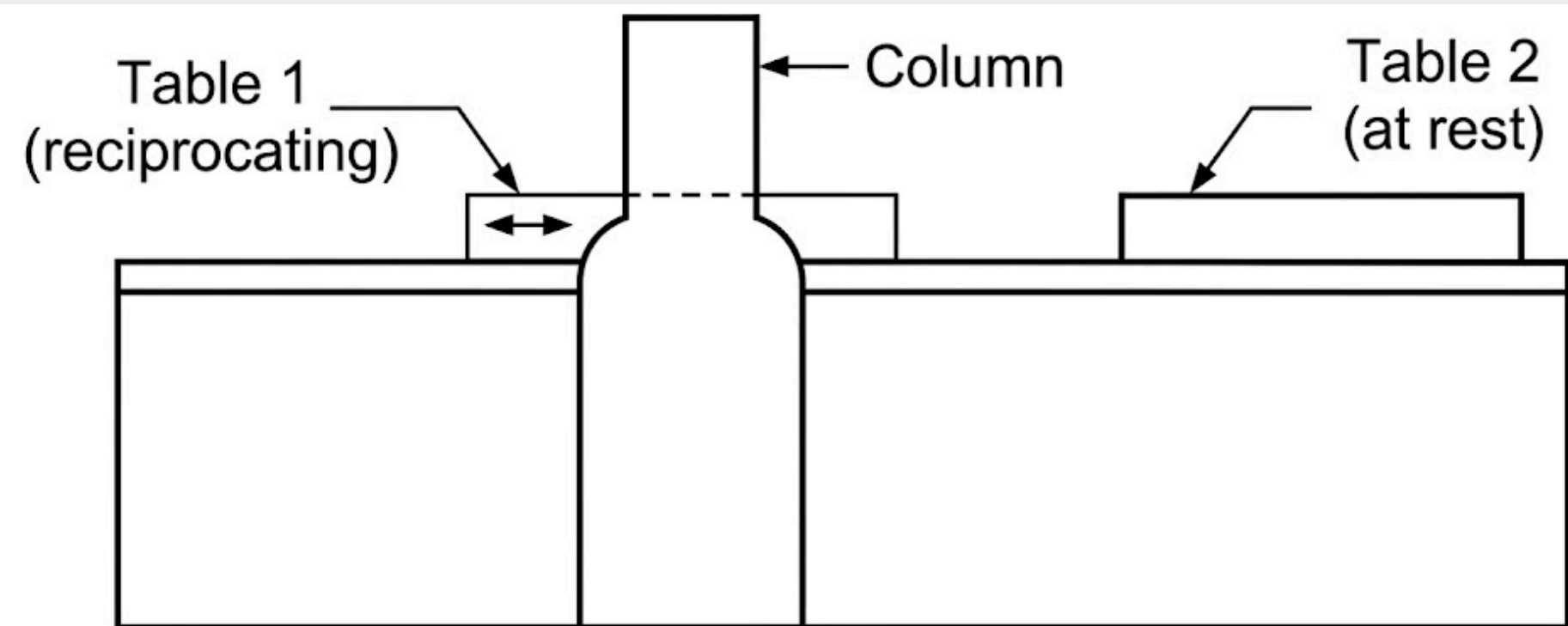


Fig. 3.34 Schematic view of a divided table planer

OPERATIONS

- **Planning flat horizontal surfaces.**
- **Planning vertical surfaces.**
- **Planning at an angle and machining dovetails.**
- **Planning curved surfaces.**
- **Planning slots and grooves.**

PLANNING FLAT HORIZONTAL SURFACES

- **Planning Flat Horizontal Surfaces**
- **This is the most common operation where the workpiece moves linearly under a stationary single-point cutting tool. The tool is fed crosswise after each stroke to produce a perfectly level top surface.**

PLANNING VERTICAL SURFACES

- **By rotating the tool head to a vertical position and feeding the tool downward, the machine can precisely plane the sides or vertical faces of a large casting or block.**

PLANNING AT AN ANGLE OR MACHING DOVETAILS

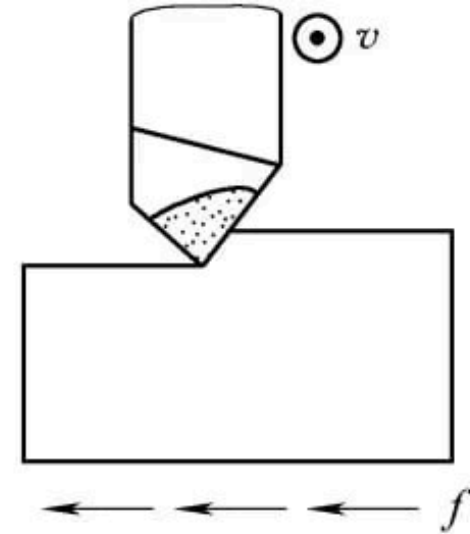
- **The tool head can be swiveled to any required angle to machine inclined surfaces. This is essential for creating dovetail joints and V-ways often found in machine tool guide-ways.**

PLANNING CURVED SURFACES

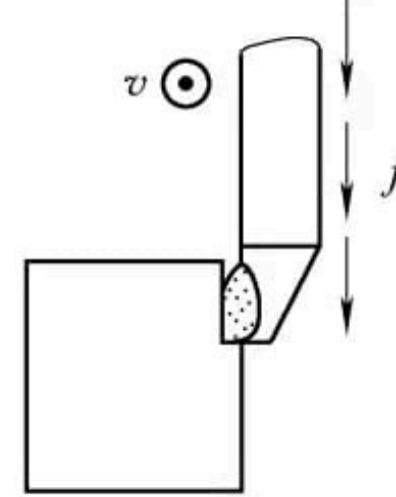
- While primarily for flat work, curved profiles can be achieved using a form tool (a tool ground to the specific shape of the curve) or by using a special contour-following attachment that guides the tool along a template.

PLANNING SLOTS AND GROOVES

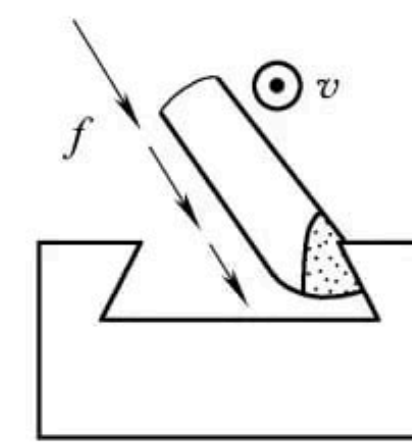
- The machine is highly effective at cutting long, straight slots, T-slots, and keyways. By using specifically shaped tools, it can machine deep grooves along the entire length of a heavy workpiece.



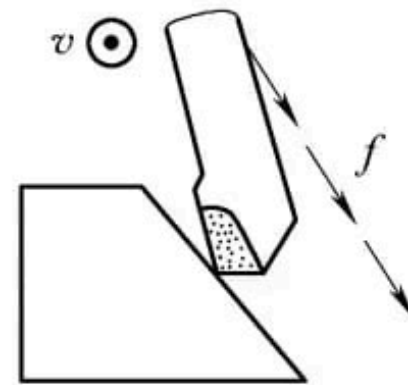
a) Flat planer blade for planing flat surfaces



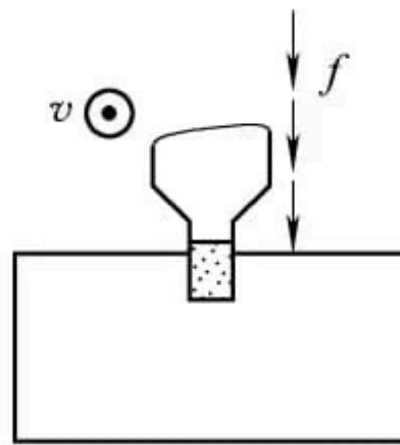
b) Offset blade for planing vertical surfaces



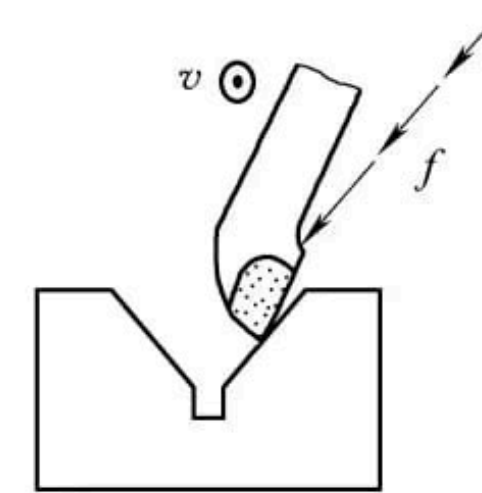
c) Angled offset blade for planing dovetail grooves



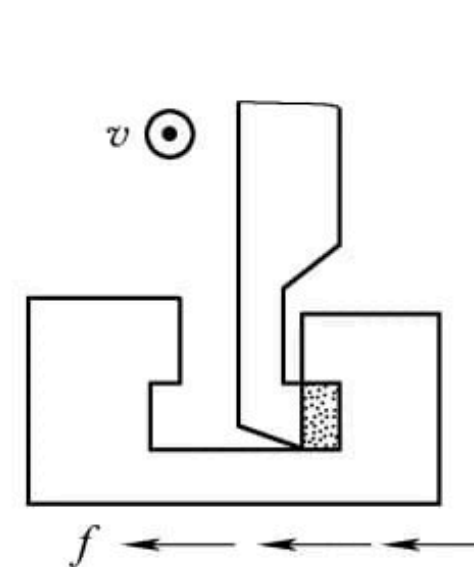
d) Offset blade for planing inclined surfaces



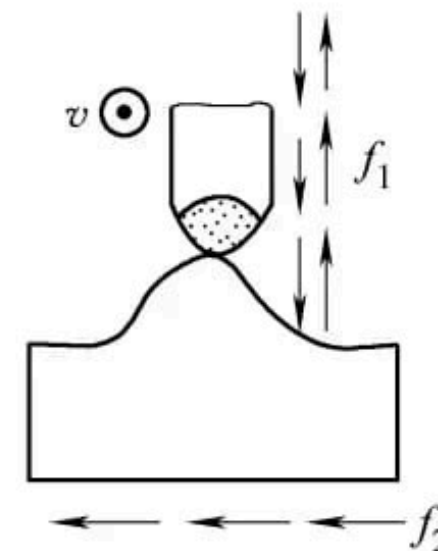
e) Cutting blade for cutting off



f) Offset blade for planing V-shaped grooves



g) Bent cutting blade for planing T-shaped grooves



h) Forming planer blade for planing contoured surfaces

PARAMETERS OF PLANAR MACHINE

- **CUTTING SPEED**

The cutting speed of a planner is the rate at the metal is removed during the forward cutting stroke.

- **FEED**

The feed in planning machine is the distance the tool head travels at the beginning of each cutting stroke expressed in mm per double stroke.

- **DEPTH OF CUT**

It is the thickness of metal removed in one cut and is measured by the perpendicular distance between the machined and no machined surface expressed in mm.

**THANK
YOU**